# UNIT 4 LEARNING

## Forms of learning

Machine learning is the branch of Artificial Intelligence that focuses on developing models and algorithms that let computers learn from data and improve from previous experience without being explicitly programmed for every task. In simple words, ML teaches the systems to think and understand like humans by learning from the data.

**Machine learning** is generally a training system to learn from past experiences and improve performance over time. Machine learning helps to predict massive amounts of data. It helps to deliver fast and accurate results to get profitable opportunities.

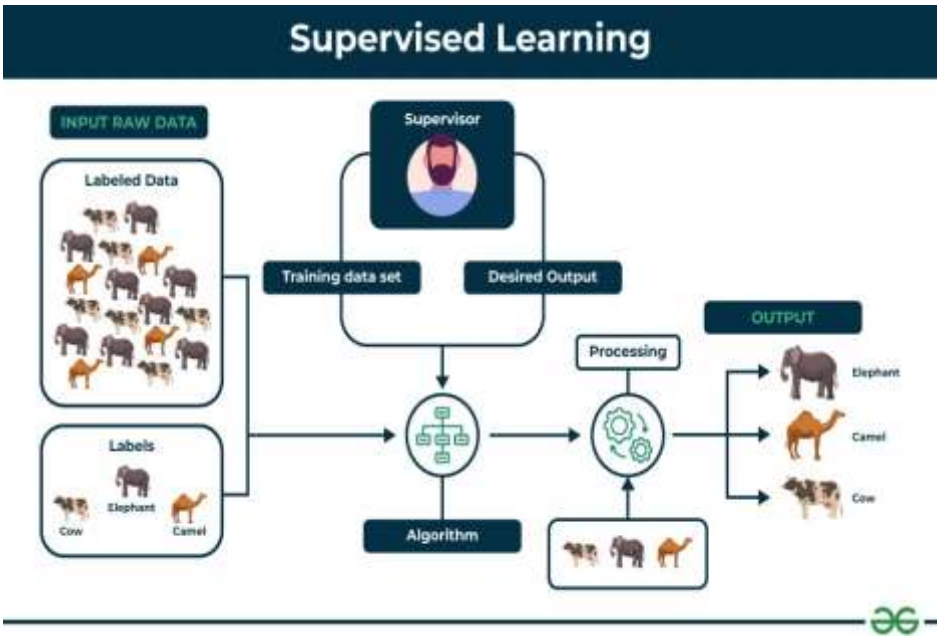
## Types of Machine Learning

There are several types of machine learning, each with special characteristics and applications. Some of the main types of machine learning algorithms are as follows:

1. Supervised Machine Learning
2. Unsupervised Machine Learning
3. Semi-Supervised Machine Learning
4. Reinforcement Learning

## 1. Supervised Machine Learning

Supervised learning is defined as when a model gets trained on a **"Labelled Dataset"**. Labelled datasets have both input and output parameters. In **Supervised Learning** algorithms learn to map points between inputs and correct outputs. It has both training and validation datasets labelled.



Let's understand it with the help of an example.

**Example:** Consider a scenario where you have to build an image classifier to differentiate between cats and dogs. If you feed the datasets of dogs and cats labelled images to the algorithm, the machine will learn to classify between a dog or a cat from these labeled images. When we input new dog or cat images that it has never seen before, it will use the learned algorithms and predict whether it is a dog or a cat. This is how **supervised learning** works, and this is particularly an image classification.

There are two main categories of supervised learning that are mentioned below:

- Classification
- Regression

*Classification*

**Classification** deals with predicting **categorical** target variables, which represent discrete classes or labels. For instance, classifying emails as spam or not spam, or predicting whether a patient has a high risk of heart disease. Classification algorithms learn to map the input features to one of the predefined classes.

Here are some classification algorithms:

- **Logistic Regression**
- **Support Vector Machine**
- **Random Forest**
- **Decision Tree**
- **K-Nearest Neighbors (KNN)**
- **Naive Bayes**

*Regression*

**Regression**, on the other hand, deals with predicting **continuous** target variables, which represent numerical values. For example, predicting the price of a house based on its size, location, and amenities, or forecasting the sales of a product. Regression algorithms learn to map the input features to a continuous numerical value.

Here are some regression algorithms:

- **Linear Regression**
- **Polynomial Regression**
- **Ridge Regression**
- **Lasso Regression**
- **Decision tree**
- **Random Forest**

**Advantages of Supervised Machine Learning**

- **Supervised Learning** models can have high accuracy as they are trained on **labelled data**.
- The process of decision-making in supervised learning models is often interpretable.
- It can often be used in pre-trained models which saves time and resources when developing new models from scratch.

**Disadvantages of Supervised Machine Learning**

- It has limitations in knowing patterns and may struggle with unseen or unexpected patterns that are not present in the training data.
- It can be time-consuming and costly as it relies on **labeled** data only.

- It may lead to poor generalizations based on new data.

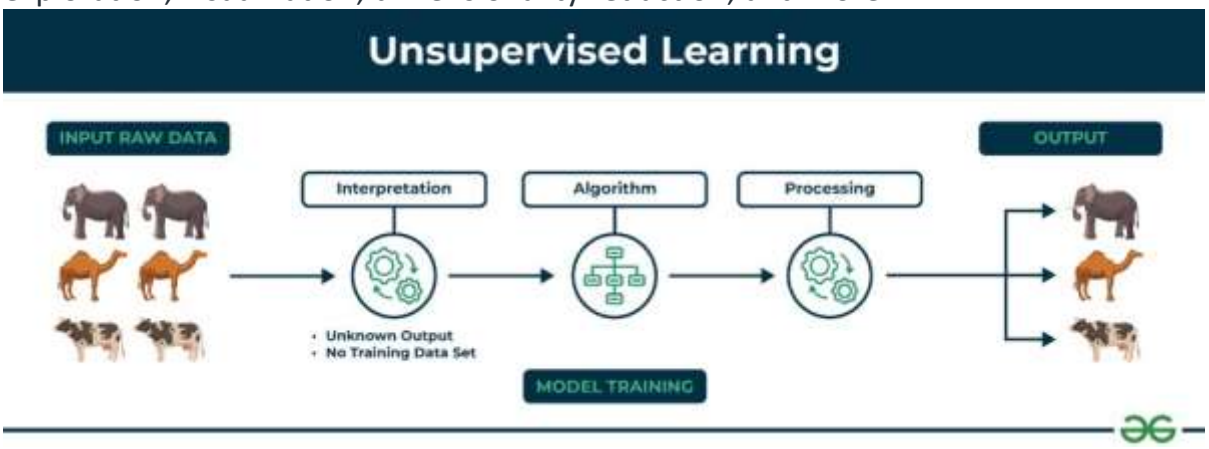
**Applications of Supervised Learning**

Supervised learning is used in a wide variety of applications, including:

- **Image classification**: Identify objects, faces, and other features in images.
- **Natural language processing:** Extract information from text, such as sentiment, entities, and relationships.
- **Speech recognition**: Convert spoken language into text.
- **Recommendation systems**: Make personalized recommendations to users.
- **Predictive analytics**: Predict outcomes, such as sales, customer churn, and stock prices.
- **Medical diagnosis**: Detect diseases and other medical conditions.
- **Fraud detection**: Identify fraudulent transactions.
- **Autonomous vehicles**: Recognize and respond to objects in the environment.
- **Email spam detection**: Classify emails as spam or not spam.
- **Quality control in manufacturing**: Inspect products for defects.
- **Credit scoring**: Assess the risk of a borrower defaulting on a loan.
- **Gaming**: Recognize characters, analyze player behavior, and create NPCs.
- **Customer support**: Automate customer support tasks.
- **Weather forecasting**: Make predictions for temperature, precipitation, and other meteorological parameters.
- **Sports analytics**: Analyze player performance, make game predictions, and optimize strategies.

## 2. Unsupervised Machine Learning

Unsupervised Learning Unsupervised learning is a type of machine learning technique in which an algorithm discovers patterns and relationships using unlabeled data. Unlike supervised learning, unsupervised learning doesn't involve providing the algorithm with labeled target outputs. The primary goal of Unsupervised learning is often to discover hidden patterns, similarities, or clusters within the data, which can then be used for various purposes, such as data exploration, visualization, dimensionality reduction, and more.



Let's understand it with the help of an example.

**Example:** Consider that you have a dataset that contains information about the purchases you made from the shop. Through clustering, the algorithm can group the same purchasing behavior among you and other customers, which reveals potential customers without predefined labels. This type of information can help businesses get target customers as well as identify outliers.

There are two main categories of unsupervised learning that are mentioned below:

- Clustering
- Association

**Clustering**

Clustering is the process of grouping data points into clusters based on their similarity. This technique is useful for identifying patterns and relationships in data without the need for labeled examples.

Here are some clustering algorithms:

- **K-Means Clustering algorithm**
- **Mean-shift algorithm**
- **DBSCAN Algorithm**
- **Principal Component Analysis**
- **Independent Component Analysis**

**Association**

Association rule learning is a technique for discovering relationships between items in a dataset. It identifies rules that indicate the presence of one item implies the presence of another item with a specific probability.

Here are some association rule learning algorithms:

- **Apriori Algorithm**
- **Eclat**
- **FP-growth Algorithm**

**Advantages of Unsupervised Machine Learning**

- It helps to discover hidden patterns and various relationships between the data.
- Used for tasks such as **customer segmentation, anomaly detection,** and **data exploration**.
- It does not require labeled data and reduces the effort of data labeling.

**Disadvantages of Unsupervised Machine Learning**

- Without using labels, it may be difficult to predict the quality of the model's output.

- Cluster Interpretability may not be clear and may not have meaningful interpretations.
- It has techniques such as autoencoders and dimensionality reduction that can be used to extract meaningful features from raw data.

**Applications of Unsupervised Learning**

Here are some common applications of unsupervised learning:

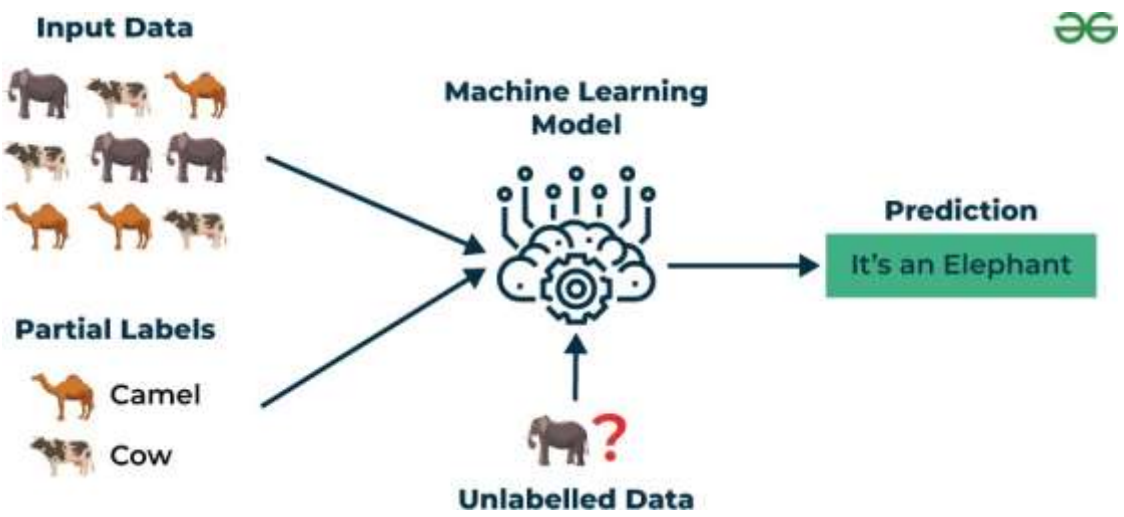
- **Clustering**: Group similar data points into clusters.
- **Anomaly detection**: Identify outliers or anomalies in data.
- **Dimensionality reduction**: Reduce the dimensionality of data while preserving its essential information.
- **Recommendation systems**: Suggest products, movies, or content to users based on their historical behavior or preferences.
- **Topic modeling**: Discover latent topics within a collection of documents.
- **Density estimation**: Estimate the probability density function of data.
- **Image and video compression**: Reduce the amount of storage required for multimedia content.
- **Data preprocessing**: Help with data preprocessing tasks such as data cleaning, imputation of missing values, and data scaling.
- **Market basket analysis**: Discover associations between products.
- **Genomic data analysis**: Identify patterns or group genes with similar expression profiles.
- **Image segmentation**: Segment images into meaningful regions.
- **Community detection in social networks**: Identify communities or groups of individuals with similar interests or connections.
- **Customer behavior analysis**: Uncover patterns and insights for better marketing and product recommendations.
- **Content recommendation**: Classify and tag content to make it easier to recommend similar items to users.
- **Exploratory data analysis (EDA)**: Explore data and gain insights before defining specific tasks.

**3. Semi-Supervised Learning**

Semi-Supervised learning is a machine learning algorithm that works between the supervised and unsupervised learning so it uses both **labelled and unlabelled** data. It's particularly useful when obtaining labeled data is costly, time-consuming, or resource-intensive. This approach is useful when the dataset is

expensive and time-consuming. Semi-supervised learning is chosen when labeled data requires skills and relevant resources in order to train or learn from it.

We use these techniques when we are dealing with data that is a little bit labeled and the rest large portion of it is unlabeled. We can use the unsupervised techniques to predict labels and then feed these labels to supervised techniques. This technique is mostly applicable in the case of image data sets where usually all images are not labeled.




Let's understand it with the help of an example.

**Example**: Consider that we are building a language translation model, having labeled translations for every sentence pair can be resources intensive. It allows the models to learn from labeled and unlabeled sentence pairs, making them more accurate. This technique has led to significant improvements in the quality of machine translation services.

*Types of Semi-Supervised Learning Methods*

There are a number of different semi-supervised learning methods each with its own characteristics. Some of the most common ones include:

- **Graph-based semi-supervised learning:** This approach uses a graph to represent the relationships between the data points. The graph is then used to propagate labels from the labeled data points to the unlabeled data points.
    - **Label propagation:** This approach iteratively propagates labels from the labeled data points to the unlabeled data points, based on the similarities between the data points.

- **Co-training:** This approach trains two different machine learning models on different subsets of the unlabeled data. The two models are then used to label each other's predictions.
- **Self-training:** This approach trains a machine learning model on the labeled data and then uses the model to predict labels for the unlabeled data. The model is then retrained on the labeled data and the predicted labels for the unlabeled data.
- **Generative adversarial networks (GANs):** GANs are a type of deep learning algorithm that can be used to generate synthetic data. GANs can be used to generate unlabeled data for semi-supervised learning by training two neural networks, a generator and a discriminator.

*Advantages of Semi- Supervised Machine Learning*

- It leads to better generalization as compared to **supervised learning,** as it takes both labeled and unlabeled data.
- Can be applied to a wide range of data.

*Disadvantages of Semi- Supervised Machine Learning*

- **Semi-supervised** methods can be more complex to implement compared to other approaches.
- It still requires some **labeled data** that might not always be available or easy to obtain.
- The unlabeled data can impact the model performance accordingly.

*Applications of Semi-Supervised Learning*

Here are some common applications of semi-supervised learning:

- **Image Classification and Object Recognition**: Improve the accuracy of models by combining a small set of labeled images with a larger set of unlabeled images.
- **Natural Language Processing (NLP)**: Enhance the performance of language models and classifiers by combining a small set of labeled text data with a vast amount of unlabeled text.
- **Speech Recognition:** Improve the accuracy of speech recognition by leveraging a limited amount of transcribed speech data and a more extensive set of unlabeled audio.
- **Recommendation Systems**: Improve the accuracy of personalized recommendations by supplementing a sparse set of user-item interactions (labeled data) with a wealth of unlabeled user behavior data.
- **Healthcare and Medical Imaging**: Enhance medical image analysis by utilizing a small set of labeled medical images alongside a larger set of unlabeled images.
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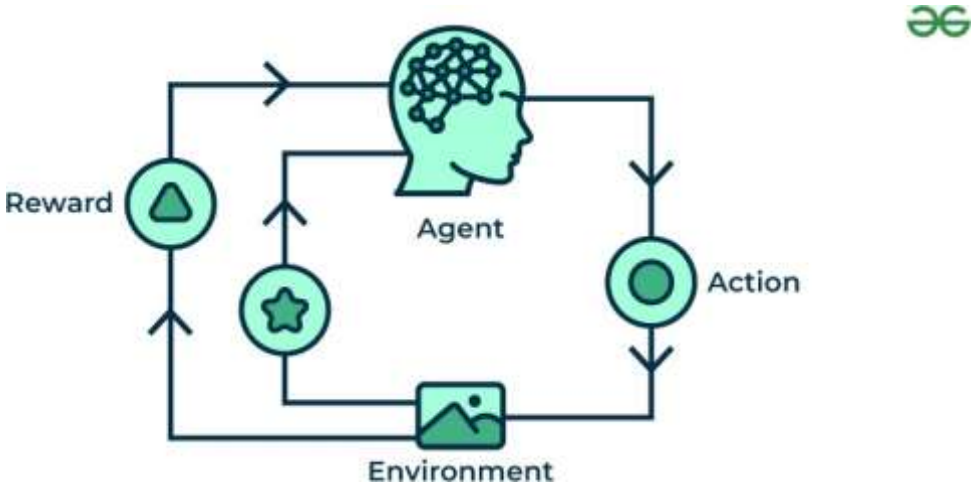
## 4. Reinforcement Machine Learning

AlphReinforcement machine learning algorithm is a learning method that interacts with the environment by producing actions and discovering errors. **Trial, error, and delay** are the most relevant characteristics of reinforcement learning. In this technique, the model keeps on increasing its performance using Reward Feedback to learn the behavior or pattern. These algorithms are specific to a particular problem e.g. Google Self Driving car, aGo where a bot competes with humans and even itself to get better and better performers in Go Game. Each time we feed in data, they learn and add the data to their

knowledge which is training data. So, the more it learns the better it gets trained and hence experienced.

Here are some of most common reinforcement learning algorithms:

- **Q-learning:** Q-learning is a model-free RL algorithm that learns a Q-function, which maps states to actions. The Q-function estimates the expected reward of taking a particular action in a given state.
- **SARSA (State-Action-Reward-State-Action):** SARSA is another model-free RL algorithm that learns a Q-function. However, unlike Q-learning, SARSA updates the Q-function for the action that was actually taken, rather than the optimal action.
- **Deep Q-learning:** Deep Q-learning is a combination of Q-learning and deep learning. Deep Q-learning uses a neural network to represent the Q-function, which allows it to learn complex relationships between states and actions.




*Reinforcement Machine Learning*

Let's understand it with the help of examples.

**Example:** Consider that you are training an AI agent to play a game like chess. The agent explores different moves and receives positive or negative feedback based on the outcome. Reinforcement Learning also finds applications in which they learn to perform tasks by interacting with their surroundings.

*Types of Reinforcement Machine Learning*

There are two main types of reinforcement learning:

**Positive reinforcement**

- Rewards the agent for taking a desired action.
- Encourages the agent to repeat the behavior.
- Examples: Giving a treat to a dog for sitting, providing a point in a game for a correct answer.

**Negative reinforcement**

- Removes an undesirable stimulus to encourage a desired behavior.
- Discourages the agent from repeating the behavior.
- Examples: Turning off a loud buzzer when a lever is pressed, avoiding a penalty by completing a task.

**Advantages of Reinforcement Machine Learning**

- It has autonomous decision-making that is well-suited for tasks and that can learn to make a sequence of decisions, like robotics and game-playing.
- This technique is preferred to achieve long-term results that are very difficult to achieve.
- It is used to solve a complex problems that cannot be solved by conventional techniques.

**Disadvantages of Reinforcement Machine Learning**

- Training Reinforcement Learning agents can be computationally expensive and time-consuming.
- Reinforcement learning is not preferable to solving simple problems.
- It needs a lot of data and a lot of computation, which makes it impractical and costly.

**Applications of Reinforcement Machine Learning**

Here are some applications of reinforcement learning:

- **Game Playing**: RL can teach agents to play games, even complex ones.
- **Robotics**: RL can teach robots to perform tasks autonomously.
- **Autonomous Vehicles**: RL can help self-driving cars navigate and make decisions.
- **Recommendation Systems**: RL can enhance recommendation algorithms by learning user preferences.
- **Healthcare**: RL can be used to optimize treatment plans and drug discovery.
- **Natural Language Processing (NLP)**: RL can be used in dialogue systems and chatbots.
- **Finance and Trading**: RL can be used for algorithmic trading.
- **Supply Chain and Inventory Management**: RL can be used to optimize supply chain operations.
- **Energy Management**: RL can be used to optimize energy consumption.
- **Game AI**: RL can be used to create more intelligent and adaptive NPCs in video games.
- **Adaptive Personal Assistants**: RL can be used to improve personal assistants.
- **Virtual Reality (VR) and Augmented Reality (AR):** RL can be used to create immersive and interactive experiences.
- **Industrial Control**: RL can be used to optimize industrial processes.
- **Education**: RL can be used to create adaptive learning systems.
- **Agriculture**: RL can be used to optimize agricultural operations.

## Decision Tree

A decision tree in machine learning is a versatile, interpretable algorithm used for predictive modelling. It structures decisions based on input data, making it suitable for both classification and regression tasks. This article delves into the components, terminologies, construction, and advantages of decision trees, exploring their applications and learning algorithms.

# Decision Tree in Machine Learning

A decision tree is a type of supervised learning algorithm that is commonly used in machine learning to model and predict outcomes based on input data. It is a tree-like structure where each internal node tests on attribute, each branch corresponds to attribute value and each leaf node represents the final decision or prediction. The decision tree algorithm falls under the category of supervised learning. They can be used to solve both **regression** and **classification problems**.

Decision Tree Terminologies

There are specialized terms associated with decision trees that denote various components and facets of the tree structure and decision-making procedure. :

- **Root Node:** A decision tree's root node, which represents the original choice or feature from which the tree branches, is the highest node.
- **Internal Nodes (Decision Nodes)**: Nodes in the tree whose choices are determined by the values of particular attributes. There are branches on these nodes that go to other nodes.
- **Leaf Nodes (Terminal Nodes)**: The branches' termini, when choices or forecasts are decided upon. There are no more branches on leaf nodes.
- **Branches (Edges)**: Links between nodes that show how decisions are made in response to particular circumstances.
- **Splitting**: The process of dividing a node into two or more sub-nodes based on a decision criterion. It involves selecting a feature and a threshold to create subsets of data.
- **Parent Node**: A node that is split into child nodes. The original node from which a split originates.
- **Child Node**: Nodes created as a result of a split from a parent node.
- **Decision Criterion**: The rule or condition used to determine how the data should be split at a decision node. It involves comparing feature values against a threshold.
- **Pruning**: The process of removing branches or nodes from a decision tree to improve its generalisation and prevent overfitting.

**How Decision Tree is formed?**

The process of forming a decision tree involves recursively partitioning the data based on the values of different attributes. The algorithm selects the best attribute to split the data at each internal node, based on certain criteria such as information gain or Gini impurity. This splitting process continues until a stopping criterion is met, such as reaching a maximum depth or having a minimum number of instances in a leaf node.

**Why Decision Tree?**

Decision trees are widely used in machine learning for a number of reasons:

- Decision trees are so versatile in simulating intricate decision-making processes, because of their interpretability and versatility.
- Their portrayal of complex choice scenarios that take into account a variety of causes and outcomes is made possible by their hierarchical structure.
- They provide comprehensible insights into the decision logic, decision trees are especially helpful for tasks involving categorisation and regression.

- They are proficient with both numerical and categorical data, and they can easily adapt to a variety of datasets thanks to their autonomous feature selection capability.
- Decision trees also provide simple visualization, which helps to comprehend and elucidate the underlying decision processes in a model.

We have two popular attribute selection measures:

1. Information Gain
2. Gini Index

**1. Information Gain:**

When we use a node in a decision tree to partition the training instances into smaller subsets the entropy changes. Information gain is a measure of this change in entropy.

- Suppose S is a set of instances,
- A is an attribute
- Sv is the subset of S
- *v* represents an individual value that the attribute *A* can take and Values (A) is the set of all possible values of A, then

**Entropy:** is the measure of uncertainty of a random variable, it characterizes the impurity of an arbitrary collection of examples. The higher the entropy more the information content.

Suppose S is a set of instances, A is an attribute, Sv is the subset of S with A = v, and Values (A) is the set of all possible values of A, then

$\sum Gain(S,A)=Entropy(S)-\sum v\epsilon Values(A)|S||Sv|.Entropy(Sv)$

Example:

```
For the set X = {a,a,a,b,b,b,b,b}
Total instances: 8
Instances of b: 5
Instances of a: 3
```

$$\begin{aligned}\text{Entropy } H(X) &= \left[\left(\frac{3}{8}\right)\log_2\frac{3}{8}+\left(\frac{5}{8}\right)\log_2\frac{5}{8}\right]\\ &= -[0.375(-1.415)+0.625(-0.678)]\\ &= -(-0.53-0.424)\\ &= 0.954\end{aligned}$$

## Regression and Classification with Linear Models

**What is Regression?**

Regression is a statistical approach used to analyze the relationship between a dependent variable (target variable) and one or more independent variables (predictor variables). The

objective is to determine the most suitable function that characterizes the connection between these variables.

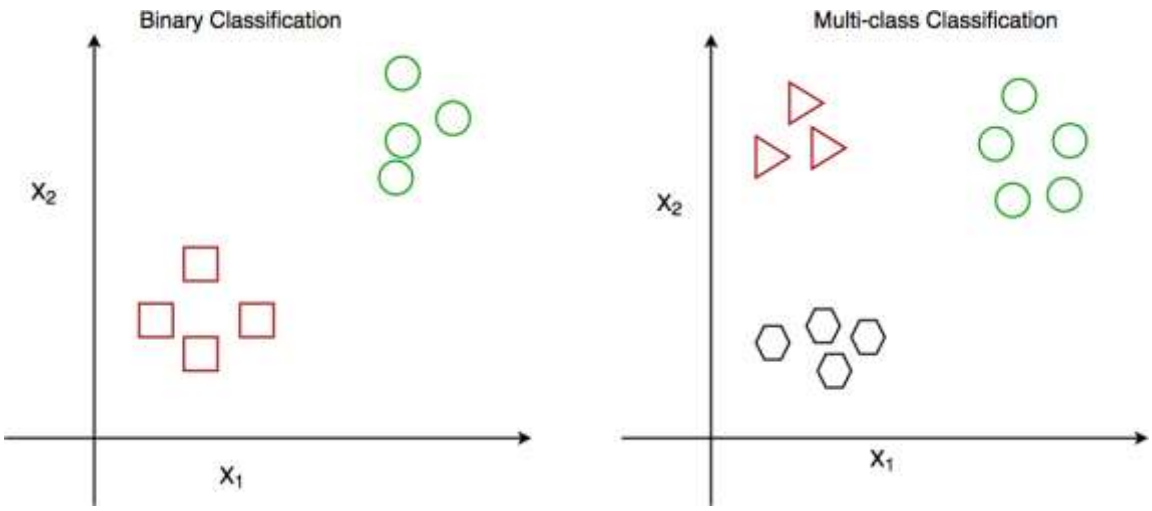
Classification and Regression are two major prediction problems that are usually dealt with in Data Mining and Machine Learning. We are going to deal with both Classification and Regression and we will also see differences between them in this article.

**Classification Algorithms**

**Classification** is the process of finding or discovering a model or function that helps in separating the data into multiple categorical classes i.e. discrete values. In classification, data is categorized under different labels according to some parameters given in the input and then the labels are predicted for the data.

- In a classification task, we are supposed to predict discrete target variables(class labels) using independent features.
- In the classification task, we are supposed to find a decision boundary that can separate the different classes in the target variable.

The derived mapping function could be demonstrated in the form of "IF-THEN" rules. The classification process deals with problems where the data can be divided into binary or multiple discrete labels. Let's take an example, suppose we want to predict the possibility of the winning of a match by Team A on the basis of some parameters recorded earlier. Then there would be two labels Yes and No.




Types of Classification Algorithms

There are different types of State of the art classification algorithms that have been developed over time to give the best results for classification tasks by employing techniques like bagging and boosting.

- Decision Tree
- Random Forest Classifier
- K – Nearest Neighbors
- Support Vector Machine

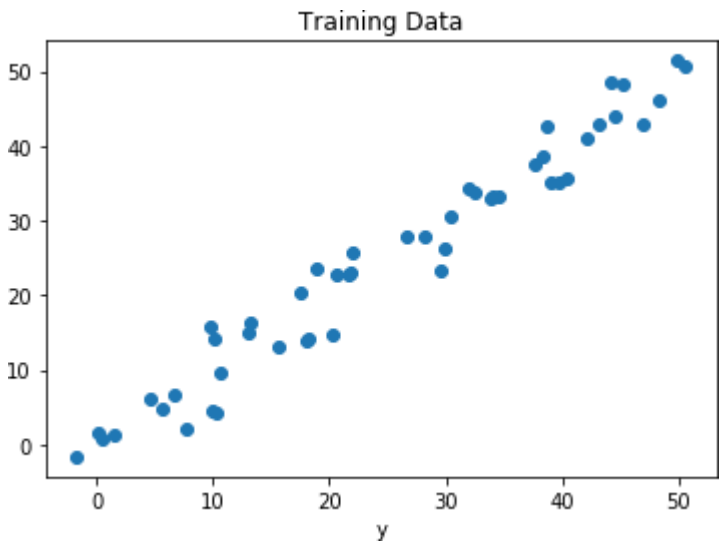
**Regression Algorithms**

**Regression** is the process of finding a model or function for distinguishing the data into continuous real values instead of using classes or discrete values. It can also identify the distribution movement depending on the historical data. Because a regression predictive

model predicts a quantity, therefore, the skill of the model must be reported as an error in those predictions.

- In a regression task, we are supposed to predict a continuous target variable using independent features.
- In the regression tasks, we are faced with generally two types of problems linear and non-linear regression.

Let's take a similar example in regression also, where we are finding the possibility of rain in some particular regions with the help of some parameters recorded earlier. Then there is a probability associated with the rain.




## Types of Regression Algorithms

There are different types of State of the art regression algorithms that have been developed over time to give the best results for regression tasks by employing techniques like bagging and boosting.

- Lasso Regression
- Ridge Regression
- XGBoost Regressor
- LGBM Regressor

**Comparison between Classification and Regression**

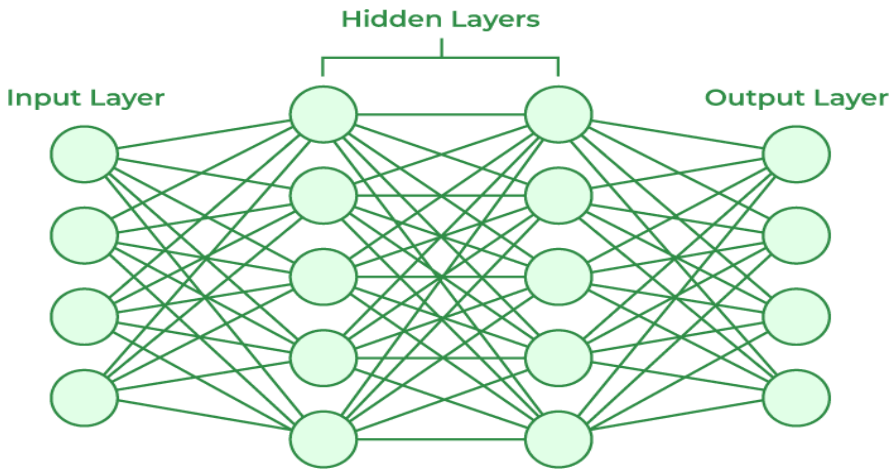
| Classification | Regression |
|---|---|
| In this problem statement, the target variables are discrete. | In this problem statement, the target variables are continuous. |
| Problems like Spam Email Classification, Disease prediction like | Problems like House Price Prediction, Rainfall Prediction like |

| Classification | Regression |
|---|---|
| problems are solved using Classification Algorithms. | problems are solved using regression Algorithms. |
| In this algorithm, we try to find the best possible decision boundary which can separate the two classes with the maximum possible separation. | In this algorithm, we try to find the best-fit line which can represent the overall trend in the data. |
| Evaluation metrics like Precision, Recall, and F1-Score are used here to evaluate the performance of the classification algorithms. | Evaluation metrics like Mean Squared Error, R2-Score, and MAPE are used here to evaluate the performance of the regression algorithms. |
| Here we face the problems like binary Classification or Multi-Class Classification problems. | Here we face the problems like Linear Regression models as well as non-linear models. |
| Input Data are Independent variables and categorical dependent variable. | Input Data are Independent variables and continuous dependent variable. |
| The classification algorithm's task mapping the input value of x with the discrete output variable of y. | The regression algorithm's task is mapping input value (x) with continuous output variable (y). |
| Output is Categorical labels. | Output is Continuous numerical values. |
| Objective is to Predict categorical/class labels. | Objective is to Predicting continuous numerical values. |
| Example use cases are Spam detection, image recognition, sentiment analysis | Example use cases are Stock price prediction, house price prediction, demand forecasting. |

## Artificial neuron network

- Artificial Neural Networks contain artificial neurons which are called **units**.
- These units are arranged in a series of layers that together constitute the whole Artificial Neural Network in a system.
- A layer can have only a dozen units or millions of units as this depends on how the complex neural networks will be required to learn the hidden patterns in the dataset.
- Commonly, Artificial Neural Network has an input layer, an output layer as well as hidden layers. The input layer receives data from the outside world which the neural network needs to analyze or learn about.
- Then this data passes through one or multiple hidden layers that transform the input into data that is valuable for the output layer. Finally, the output layer provides an output in the form of a response of the Artificial Neural Networks to input data provided.

In the majority of neural networks, units are interconnected from one layer to another. Each of these connections has weights that determine the influence of one unit on another unit. As the data transfers from one unit to another, the neural network learns more and more about the data which eventually results in an output from the output layer.




The structures and operations of human neurons serve as the basis for artificial neural networks. It is also known as neural networks or neural nets. The input layer of an artificial neural network is the first layer, and it receives input from external sources and releases it to the hidden layer, which is the second layer. In the hidden layer, each neuron receives input from the previous layer neurons, computes the weighted sum, and sends it to the neurons in the next layer. These connections are weighted means effects of the inputs from the previous layer are optimized more or less by assigning different-different weights to each input and it is adjusted during the training process by optimizing these weights for improved model performance.

## types of Artificial Neural Networks?

- **Feedforward Neural Network:** The feedforward neural network is one of the most basic artificial neural networks. In this ANN, the

data or the input provided travels in a single direction. It enters into the ANN through the input layer and exits through the output layer while hidden layers may or may not exist. So the feedforward neural network has a front-propagated wave only and usually does not have backpropagation.

- **Convolutional Neural Network:** A Convolutional neural network has some similarities to the feed-forward neural network, where the connections between units have weights that determine the influence of one unit on another unit. But a CNN has one or more than one convolutional layer that uses a convolution operation on the input and then passes the result obtained in the form of output to the next layer. CNN has applications in speech and image processing which is particularly useful in computer vision.
- **Modular Neural Network:** A Modular Neural Network contains a collection of different neural networks that work independently towards obtaining the output with no interaction between them. Each of the different neural networks performs a different sub-task by obtaining unique inputs compared to other networks. The advantage of this modular neural network is that it breaks down a large and complex computational process into smaller components, thus decreasing its complexity while still obtaining the required output.
- **Radial basis function Neural Network:** Radial basis functions are those functions that consider the distance of a point concerning the center. RBF functions have two layers. In the first layer, the input is mapped into all the Radial basis functions in the hidden layer and then the output layer computes the output in the next step. Radial basis function nets are normally used to model the data that represents any underlying trend or function.
- **Recurrent Neural Network:** The Recurrent Neural Network saves the output of a layer and feeds this output back to the input to better predict the outcome of the layer. The first layer in the RNN is quite similar to the feed-forward neural network and the recurrent neural network starts once the output of the first layer is computed. After this layer, each unit will remember some information from the previous step so that it can act as a memory cell in performing computations.

## Applications of Artificial Neural Networks

1. **Social Media:** Artificial Neural Networks are used heavily in Social Media. For example, let's take the **'People you may know'** feature on Facebook that suggests people that you might know in real life so that you can send them friend requests. Well, this magical effect is achieved by using Artificial Neural Networks that analyze your profile, your interests, your current friends, and also their friends and various other factors to calculate the people you might potentially know. Another common application of Machine Learning in social media is **facial recognition**. This is done by finding around 100 reference points on the person's face and then matching them with those already available in the database using convolutional neural networks.
2. **Marketing and Sales:** When you log onto E-commerce sites like Amazon and Flipkart, they will recommend your products to buy based on your previous browsing history. Similarly, suppose you love Pasta, then Zomato, Swiggy, etc. will show you restaurant recommendations based on your tastes and previous order history. This is true across all new-age marketing segments like Book sites, Movie services, Hospitality sites, etc. and it is done by implementing **personalized marketing**. This uses Artificial Neural Networks to identify the customer likes, dislikes, previous shopping history, etc., and then tailor the marketing campaigns accordingly.
3. **Healthcare**: Artificial Neural Networks are used in Oncology to train algorithms that can identify cancerous tissue at the microscopic level at the same accuracy as trained physicians. Various rare diseases may manifest in physical characteristics and can be identified in their premature stages by using **Facial Analysis** on the patient photos. So the full-scale implementation of Artificial Neural Networks in the healthcare environment can only enhance the diagnostic abilities of medical experts and ultimately lead to the overall improvement in the quality of medical care all over the world.
4. **Personal Assistants:** I am sure you all have heard of Siri, Alexa, Cortana, etc., and also heard them based on the phones you have!!! These are personal assistants and an example of speech recognition that uses **Natural Language Processing** to interact with the users and formulate a response accordingly. Natural Language Processing uses artificial neural networks that are made to handle many tasks of these personal assistants such as managing the language syntax, semantics, correct speech, the conversation that is going on, etc.

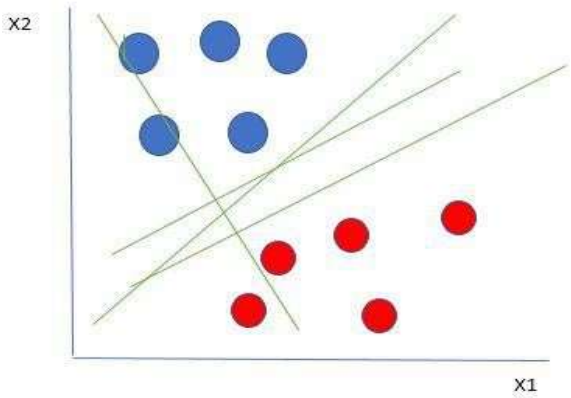
## Support vector machine

### Support Vector Machine

Support Vector Machine (SVM) is a supervised machine learning algorithm used for both classification and regression. Though we say regression problems as well it's best suited for classification. The main objective of the SVM algorithm is to find the optimal hyperplane in an N-dimensional space that can separate the data points in different classes in the feature space. The hyperplane tries that the margin between the closest points of different classes should be as maximum as possible. The dimension of the hyperplane depends upon the number of features. If the number of input features is two, then the hyperplane is just a line. If the number of input features is three, then the hyperplane

becomes a 2-D plane. It becomes difficult to imagine when the number of features exceeds three.

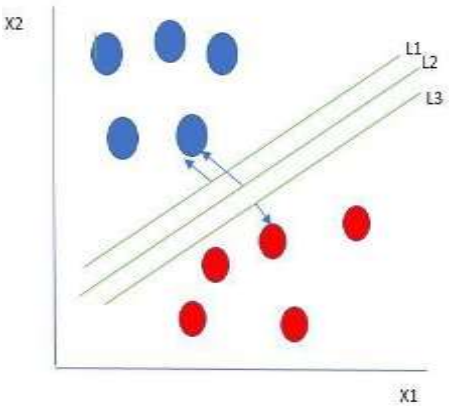
let's consider two independent variables $x_1$, $x_2$, and one dependent variable which is either a blue circle or a red circle.



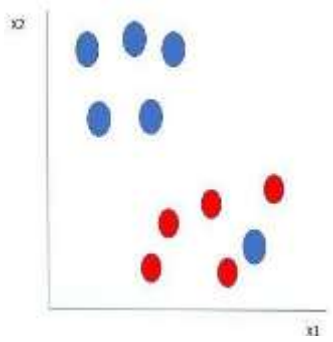

From the figure above it's very clear that there are multiple lines (our hyperplane here is a line because we are considering only two input features $x_1$, $x_2$) that segregate our data points or do a classification between red and blue circles. So how do we choose the best line or in general the best hyperplane that segregates our data points?

How does SVM work?

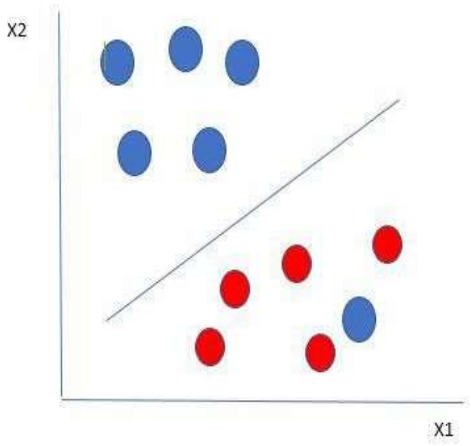
One reasonable choice as the best hyperplane is the one that represents the largest separation or margin between the two classes.




So we choose the hyperplane whose distance from it to the nearest data point on each side is maximized. If such a hyperplane exists it is known as the **maximum-margin hyperplane/hard margin**. So from the above figure, we choose L2. Let's consider a scenario like shown below

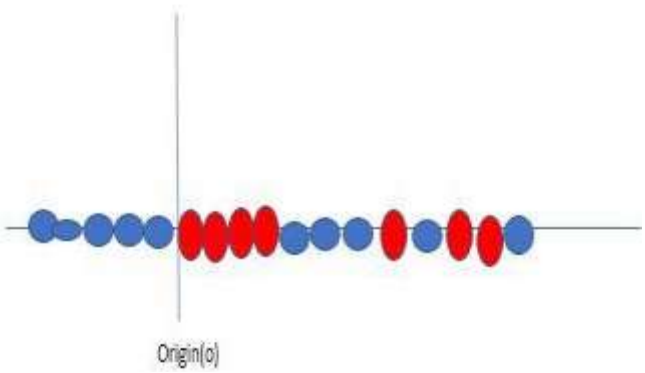



Here we have one blue ball in the boundary of the red ball. So how does SVM classify the data? It's simple! The blue ball in the boundary of red ones is an outlier of blue balls. The SVM algorithm has the characteristics to ignore the outlier and finds the best hyperplane that maximizes the margin. SVM is robust to outliers.

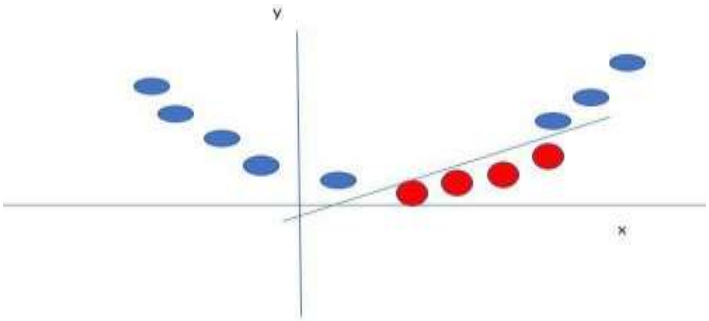



So in this type of data point what SVM does is, finds the maximum margin as done with previous data sets along with that it adds a penalty each time a point crosses the margin. So the margins in these types of cases are called **soft margins**. When there is a soft margin to the data set, the SVM tries to minimize *(1/margin+*A*(∑penalty))*. Hinge loss is a commonly used penalty. If no violations no hinge loss.If violations hinge loss proportional to the distance of violation.

Till now, we were talking about linearly separable data(the group of blue balls and red balls are separable by a straight line/linear line). What to do if data are not linearly separable?



say, our data is shown in the figure above. SVM solves this by creating a new variable using a **kernel**. We call a point $x_i$ on the line and we create a new variable $y_i$ as a function of distance from origin o.so if we plot this we get something like as shown below




In this case, the new variable y is created as a function of distance from the origin. A non-linear function that creates a new variable is referred to as a kernel.

Support Vector Machine Terminology

1. **Hyperplane:** Hyperplane is the decision boundary that is used to separate the data points of different classes in a feature space. In the case of linear classifications, it will be a linear equation i.e. wx+b = 0.
2. **Support Vectors:** Support vectors are the closest data points to the hyperplane, which makes a critical role in deciding the hyperplane and margin.
3. **Margin**: Margin is the distance between the support vector and hyperplane. The main objective of the support vector machine algorithm is to maximize the margin. The wider margin indicates better classification performance.
4. **Kernel**: Kernel is the mathematical function, which is used in SVM to map the original input data points into high-dimensional feature spaces, so, that the hyperplane can be easily found out even if the data points are not linearly separable in the original input space. Some of the common kernel functions are linear, polynomial, radial basis function(RBF), and sigmoid.
5. **Hard Margin:** The maximum-margin hyperplane or the hard margin hyperplane is a hyperplane that properly separates the data points of different categories without any misclassifications.
6. **Soft Margin:** When the data is not perfectly separable or contains outliers, SVM permits a soft margin technique. Each data point has a slack variable introduced by the soft-margin SVM formulation, which softens the strict margin requirement and permits certain misclassifications or violations. It discovers a compromise between increasing the margin and reducing violations.
7. **C:** Margin maximisation and misclassification fines are balanced by the regularisation parameter C in SVM. The penalty for going over the margin or misclassifying data items is decided by it. A stricter penalty is imposed with a greater value of C, which results in a smaller margin and perhaps fewer misclassifications.

8. **Hinge Loss:** A typical loss function in SVMs is hinge loss. It punishes incorrect classifications or margin violations. The objective function in SVM is frequently formed by combining it with the regularisation term.
9. **Dual Problem:** A dual Problem of the optimisation problem that requires locating the Lagrange multipliers related to the support vectors can be used to solve SVM. The dual formulation enables the use of kernel tricks and more effective computing.

Types of Support Vector Machine

Based on the nature of the decision boundary, Support Vector Machines (SVM) can be divided into two main parts:

- **Linear SVM:** Linear SVMs use a linear decision boundary to separate the data points of different classes. When the data can be precisely linearly separated, linear SVMs are very suitable. This means that a single straight line (in 2D) or a hyperplane (in higher dimensions) can entirely divide the data points into their respective classes. A hyperplane that maximizes the margin between the classes is the decision boundary.
- **Non-Linear SVM:** Non-Linear SVM can be used to classify data when it cannot be separated into two classes by a straight line (in the case of 2D). By using kernel functions, nonlinear SVMs can handle nonlinearly separable data. The original input data is transformed by these kernel functions into a higher-dimensional feature space, where the data points can be linearly separated. A linear SVM is used to locate a nonlinear decision boundary in this modified space.

**Advantages of SVM**

- Effective in high-dimensional cases.
- Its memory is efficient as it uses a subset of training points in the decision function called support vectors.
- Different kernel functions can be specified for the decision functions and its possible to specify custom kernels.